



# Solving Proportions

You can use proportions to solve many kinds of problems. Use cross products to solve a proportion.

To solve using a proportion, be sure that the two ratios have corresponding units.

Example

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At Bright’s Movie Theater, 3 tickets cost \$24. How much would 4 tickets cost?

Step 1

Write a ratio with two numbers from the problem. Include labels.

$$\frac{3 \text{ tickets}}{\$24}$$

Step 2

Write a proportion, using a variable for the number you do not know.

$$\frac{3 \text{ tickets}}{\$24} = \frac{4 \text{ tickets}}{\$p}$$

$$\rightarrow (3)(p) = (4)(24) \rightarrow 3p = 96$$
$$p = \frac{96}{3}$$
$$p = 32$$

Step 3

Multiply to get the cross products and find the unknown value.

$$3 \times p = 24 \times 4$$
$$3 \times p = 96$$
$$p = 96 \div 3$$
$$p = \$32$$

The cost of 4 tickets is **\$32**.

Another way to set up a proportion for this problem is this:  $\frac{\$24}{3 \text{ tickets}} = \frac{\$p}{4 \text{ tickets}}$

It doesn’t matter which number goes on top or bottom in the first ratio. What *does* matter is that both ratios are set up in the same order with corresponding units.

For example, in the problem above, writing the two ratios as

$$\frac{\$24}{3 \text{ tickets}} = \frac{4 \text{ tickets}}{\$p}$$

would *not* be a true proportion because they do not compare the same units in their numerators or the same units in their denominators.

Another way to express the same proportion is

$$\frac{\$24}{\$p} = \frac{3 \text{ tickets}}{4 \text{ tickets}}$$

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Lesson 6: Ratios and Proportions

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